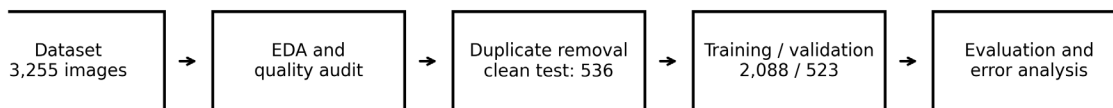


AUTOMATIC WASTE IMAGE CLASSIFICATION

From a CNN Baseline to Transfer Learning with ResNet18



Applied MSc in Data Science & AI

Deep Learning with Python - Course Project

Task: Image Classification

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Abstract

This project investigates automatic waste image classification across seven categories: cardboard, compost, glass, metal, paper, plastic, and trash. The main research question is whether transfer learning with a pre-trained ResNet18 model can improve classification performance compared with a convolutional neural network trained from scratch. A rigorous experimental workflow was applied: exploratory data analysis, preprocessing and data augmentation, stratified training/validation splitting, class-weighted learning, early stopping, model comparison, and qualitative error analysis. A data-quality audit identified 108 exact train/test duplicates in the supplied split. These test images were removed before the final evaluation, producing a leakage-free test set of 536 images. On this cleaned test set, the fine-tuned ResNet18 achieved 86.94% accuracy and 86.44% macro F1, substantially outperforming the CNN baseline (51.68% accuracy and 46.80% macro F1). The strongest remaining confusions were paper versus cardboard and plastic versus trash. The results demonstrate the value of transfer learning for small, heterogeneous image datasets while also showing why data-integrity checks are essential before interpreting model performance.

Project at a glance

Item	Value
Hugging Face task	Image Classification
Dataset	rootstrap-org/waste-classifier
Original dataset size	3,255 images (2,611 train + 644 test)
Classes	cardboard, compost, glass, metal, paper, plastic, trash
Final evaluation set	536 clean test images after exact-duplicate removal
Best model	ResNet18 fine-tuned (layer4 + classification head)
Best clean-test result	86.94% accuracy; 86.44% macro F1

Repository and notebooks

Project repository: github.com/Ali-dou/deep-learning-waste-classification

- `notebooks/01\_eda.ipynb` - exploratory data analysis and preprocessing checks.
- `notebooks/02\_cnn\_baseline.ipynb` - CNN baseline training and evaluation.
- `notebooks/03\_resnet18\_transfer\_learning.ipynb` - frozen and fine-tuned ResNet18 experiments.
- `notebooks/04\_error\_analysis.ipynb` - duplicate audit and qualitative error analysis.

1. Introduction

Reliable waste sorting is an important practical computer-vision problem. A classifier that can identify the material category of an object from an image may assist educational tools, smart bins, or manual triage workflows. However, real waste images are visually heterogeneous: they differ in lighting, object scale, background, packaging design, and context. In addition, several categories are naturally similar. A cardboard package may look like paper, while a plastic container can resemble general trash depending on its shape or contamination.

The chosen Hugging Face task is image classification. The dataset was selected because it is small enough to train locally while still containing seven meaningful categories and substantial visual variability. The task is supervised multi-class classification: each input image is assigned one mutually exclusive label from the seven available categories [1].

The project compares three strategies. First, a compact CNN is trained from scratch and serves as a baseline. Second, an ImageNet-pre-trained ResNet18 is used with its convolutional backbone frozen, so that only a new classification head learns the target categories. Third, the final residual block and classification head are fine-tuned using smaller learning rates. ResNet architectures use residual connections, which make deep networks easier to optimize and are widely used in visual recognition [2].

The scientific objective is not to maximize a leaderboard score. Instead, the project tests a clear hypothesis: pre-trained visual features should improve the classification of a relatively small waste dataset compared with learning all features from scratch. The study also treats data integrity as part of the experimental protocol: scores are interpreted only after checking for duplicate images across training and test splits.

Research question

To what extent does transfer learning with a pre-trained ResNet18 improve the robust classification of waste images compared with a CNN trained from scratch, when both models are evaluated on a test set cleaned of exact train/test duplicates?

2. Dataset and task formulation

The data were obtained from the `rootstrap-org/waste-classifier` dataset on Hugging Face [1]. The dataset card describes an image-classification dataset structured into train and test folders, with one directory per category. The supplied archive contains 3,255 images divided into 2,611 training images and 644 test images. The target classes are cardboard, compost, glass, metal, paper, plastic, and trash.

Category	Train	Provided test	Total
cardboard	346	87	433
compost	616	153	769
glass	107	27	134
metal	251	64	315
paper	452	112	564
plastic	246	62	308
trash	593	139	732
Total	2611	644	3255

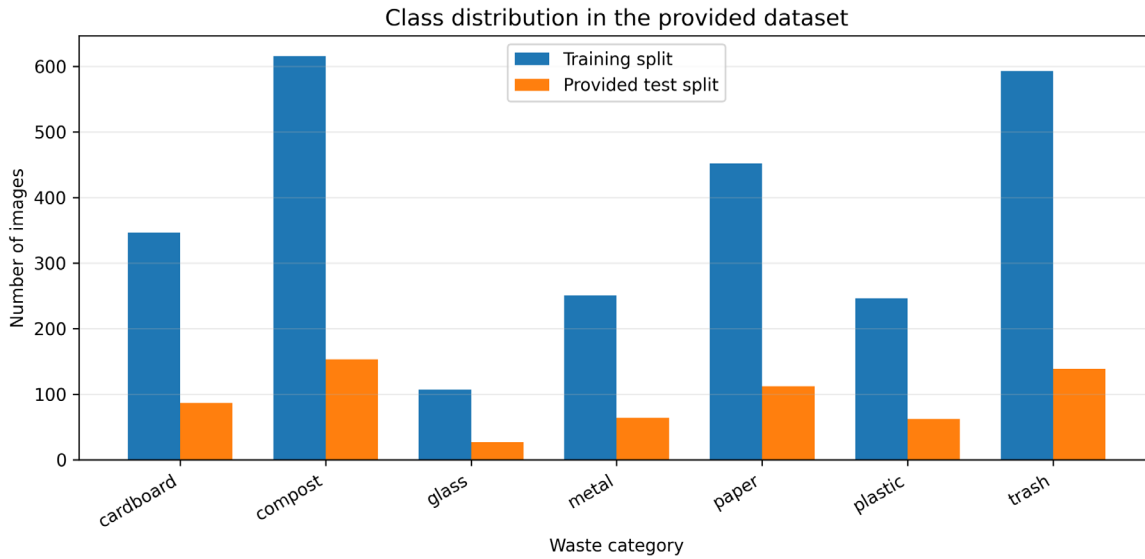


Figure 1. Class distribution in the original training and test folders.

The dataset is class-imbalanced. Compost and trash are the most represented categories in the training split, whereas glass is the smallest class. This imbalance motivates the use of macro F1 in addition to accuracy, because macro F1 gives the same weight to each category rather than allowing frequent classes to dominate the overall score.

3. Exploratory data analysis

The exploratory analysis examined class counts, random samples, and image dimensions. The visual inspection showed substantial heterogeneity: some images depict objects on a plain background or product-like stock photos, while others show discarded objects in a realistic scene. Image resolution also varies greatly. The training set has a median size close to 1000 x 990 pixels, but images range from 150 x 103 pixels to 7192 x 9508 pixels. A fixed input size is therefore necessary for batch processing.

The visual examples also reveal that the labels are not always conceptually clean. For example, a paper cup, a drug blister pack, or a composite package may contain multiple materials. In a real recycling system, the appropriate class may additionally depend on local rules. These observations are important because a model can be visually correct while disagreeing with a particular label convention.

Key EDA decisions

- Resize images to 224 x 224 pixels, enabling consistent batching and compatibility with ResNet18.
- Use training-only augmentation to reduce overfitting to object pose and orientation.
- Use class-weighted cross-entropy to reduce the influence of class imbalance.
- Report macro F1, class-level metrics, and confusion matrices in addition to accuracy.
- Perform an explicit train/test duplicate audit before retaining final results.

The EDA notebook contains the full class-distribution table, sample visualizations, and the descriptive statistics of image sizes. These outputs are preserved in `'01_eda.ipynb'` to make the workflow inspectable and reproducible.

4. Data integrity audit and leakage-free evaluation

Before accepting the model scores, an exact-duplicate audit was performed. Every image was converted to RGB and hashed from its pixel content. The hashes of all training images were compared with the hashes of all provided test images. This procedure detected 108 test images whose exact pixel content was already present in the training split. Of these duplicates, 100 had the same class label in the training data and 8 were associated with a different training label.

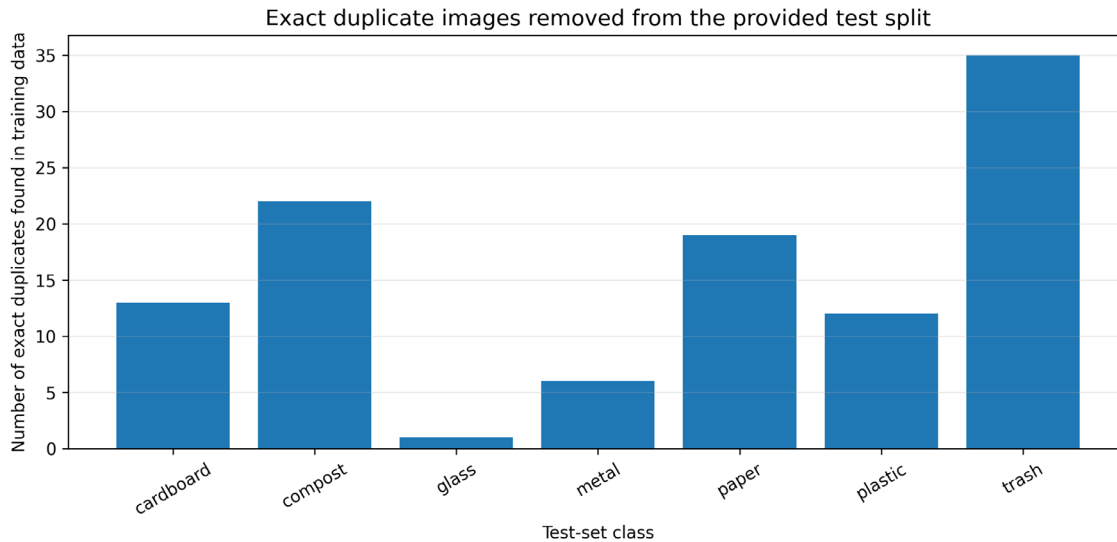


Figure 2. Exact duplicate images discovered in the provided test split, by test class.

Audit item	Result
Provided test images	644
Test images matching at least one training image exactly	108
Duplicates with at least one matching training label	100
Duplicates with only a different training label	8
Final leakage-free test images	536

The final reported metrics use only the leakage-free test set. No models were retrained because the duplicates were only found in the external test split; instead, the same saved models were re-evaluated after removing those test instances. This correction lowers the risk of artificially high scores caused by test examples that were already seen during training. It also provides a stronger basis for comparing the CNN baseline with transfer-learning approaches.

5. Preprocessing and data pipeline

All images were loaded as RGB. This step avoids issues with palette-based images or transparency and guarantees three input channels. The full provided training folder was split into training and validation subsets using a stratified 80/20 split with a fixed random seed of 42. The resulting subsets contain 2,088 training images and 523 validation images. The external test split was not used for model selection.

Split	Number of images	Purpose
Training	2,088	Parameter optimization
Validation	523	Early stopping and model selection
Provided test	644	External evaluation before integrity audit
Clean test	536	Final reported evaluation

Transformations

For the CNN baseline, training images were resized to 224 x 224, randomly flipped horizontally with probability 0.5, randomly rotated by up to 15 degrees, converted to tensors, and normalized with ImageNet mean and standard-deviation values. Validation and test images were resized to 224 x 224 and normalized without random augmentation.

For ResNet18, the training pipeline used random resized crops at 224 x 224 with a scale range of 0.75 to 1.0, horizontal flipping, and rotations up to 10 degrees. Evaluation used resizing to 256 pixels followed by a centered 224 x 224 crop. This matches the expected spatial input and normalization scheme of the ImageNet-pretrained network.

Class imbalance handling

The loss function was class-weighted cross-entropy. Class weights were computed inversely to the number of training samples per class. This does not guarantee balanced performance, but it reduces the incentive for the optimizer to obtain good global accuracy by prioritizing only the common classes.

6. Models

6.1 CNN baseline trained from scratch

The baseline is a compact CNN with four convolutional blocks. The feature extractor progressively expands the number of channels from 32 to 256. Each block contains a 3 x 3 convolution, batch normalization, ReLU activation, and max pooling, except for the final block, which ends with adaptive average pooling. The classifier consists of dropout, a 256-to-128 linear layer, ReLU, dropout, and a final seven-class linear layer. The network has 423,175 trainable parameters.

6.2 ResNet18 with frozen backbone

The second model uses ResNet18 from torchvision with default ImageNet-pretrained weights. The convolutional backbone is frozen and a new seven-class fully connected head is trained. Only 3,591 parameters are optimized in this phase. This configuration tests whether generic features learned on ImageNet can transfer to waste material categories without adapting the deep convolutional representation.

6.3 Fine-tuned ResNet18

The third model begins from the best frozen-backbone checkpoint. All parameters are frozen first, then the final residual stage ('layer4') and the classification head are unfrozen. This allows the highest-level visual features to adapt to waste-specific evidence while keeping earlier, more generic feature extractors stable. The fine-tuning configuration trains 8,397,319 parameters. Residual networks are designed to make deep feature learning easier through skip connections [2].

The study therefore isolates three levels of adaptation: learning all visual features from scratch, learning only a target head on pre-trained features, and partially adapting pre-trained features to the target domain.

7. Training setup and reproducibility

Experiment	Optimizer and learning rate	Epoch limit / patience	Selection metric
SimpleCNN	Adam, lr = 1e-3	20 epochs / 5	Validation macro F1
ResNet18 frozen	AdamW, fc lr = 1e-3, wd = 1e-4	10 epochs / 4	Validation macro F1
ResNet18 fine-tuned	AdamW, layer4 lr = 1e-5; fc lr = 1e-4; wd = 1e-4	12 epochs / 4	Validation macro F1

All experiments use a batch size of 32, weighted cross-entropy loss, a fixed random seed of 42, and early stopping based on validation macro F1. The best checkpoint of each run is saved to the `outputs/models/` directory and reloaded for test evaluation. Macro F1 was selected as the principal model-selection metric because the class distribution is uneven.

The experiments were run locally on Windows using Python 3.14.6, PyTorch 2.11.0 with CUDA 12.8, and an NVIDIA GeForce RTX 4060 Ti. PyTorch provides an imperative Python interface with GPU support, making it appropriate for this reproducible training workflow [4].

Reproducibility assets

- A `requirements.txt` file lists the Python dependencies.
- The notebooks preserve outputs, figures, metrics, and saved checkpoints.
- A fixed seed is used for data splitting and PyTorch initialization.
- The data-cleaning audit and the final clean-test index construction are retained in the notebooks.
- The README documents dataset placement, environment setup, and execution order.

8. Results

Table 5 summarizes the performance of all models on the leakage-free test set. The CNN baseline obtains 51.68% accuracy and 46.80% macro F1. Both ResNet18 variants are substantially stronger. Even when the pretrained backbone is frozen, ResNet18 reaches 85.63% accuracy and 83.89% macro F1. Fine-tuning the final residual block improves the result to 86.94% accuracy and 86.44% macro F1.

Model	Trainable parameters	Best validation macro F1	Clean-test accuracy	Clean-test macro F1
SimpleCNN from scratch	423,175	0.5376	0.5168	0.4680
ResNet18 frozen	3,591	0.8432	0.8563	0.8389
ResNet18 fine-tuned	8,397,319	0.8578	0.8694	0.8644

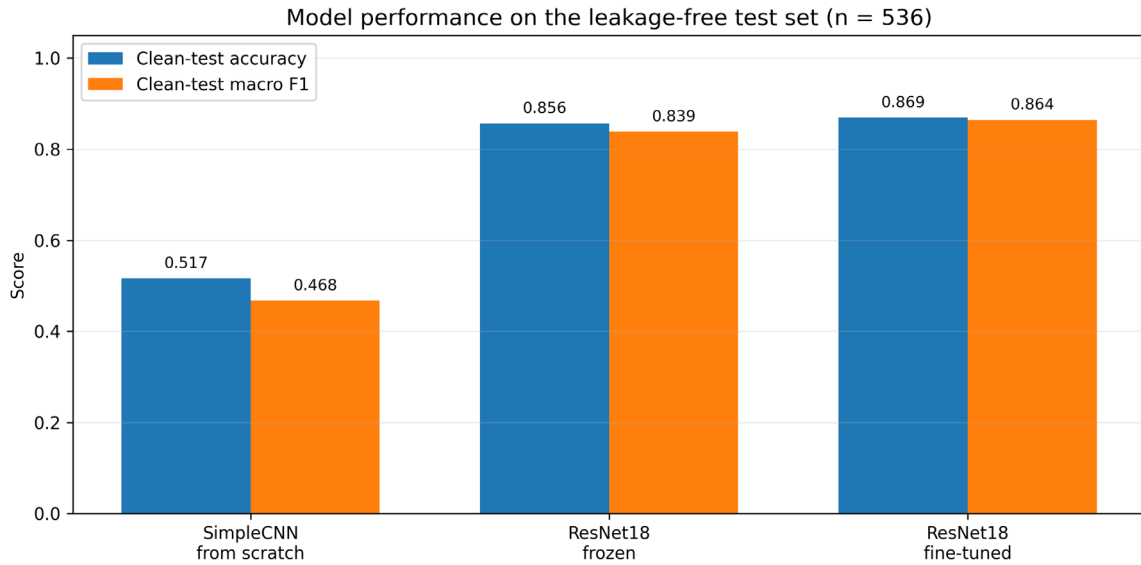


Figure 3. Model comparison on the leakage-free test set.

The results support the project hypothesis. The fine-tuned ResNet18 improves clean-test macro F1 by 39.65 percentage points over the CNN baseline. The frozen ResNet18 already captures most of the gain, which indicates that generic features from large-scale ImageNet pretraining transfer effectively to this small waste dataset. Fine-tuning produces a further 2.55-point macro-F1 gain over the frozen variant.

9. Fine-tuned ResNet18: class-level results

The fine-tuned model achieves balanced results over the seven categories. Compost and glass obtain perfect recall on the clean test set, while plastic and trash remain the most difficult categories. The macro F1 of 0.8644 is slightly below the weighted F1 of 0.8688, confirming that performance is reasonably balanced while still being affected by harder classes.

Class	Precision	Recall	F1-score	Support
cardboard	0.816	0.838	0.827	74
compost	0.970	1.000	0.985	131
glass	0.897	1.000	0.945	26
metal	0.926	0.862	0.893	58
paper	0.865	0.828	0.846	93
plastic	0.750	0.780	0.765	50
trash	0.802	0.779	0.790	104
Macro average	0.860	0.870	0.864	536

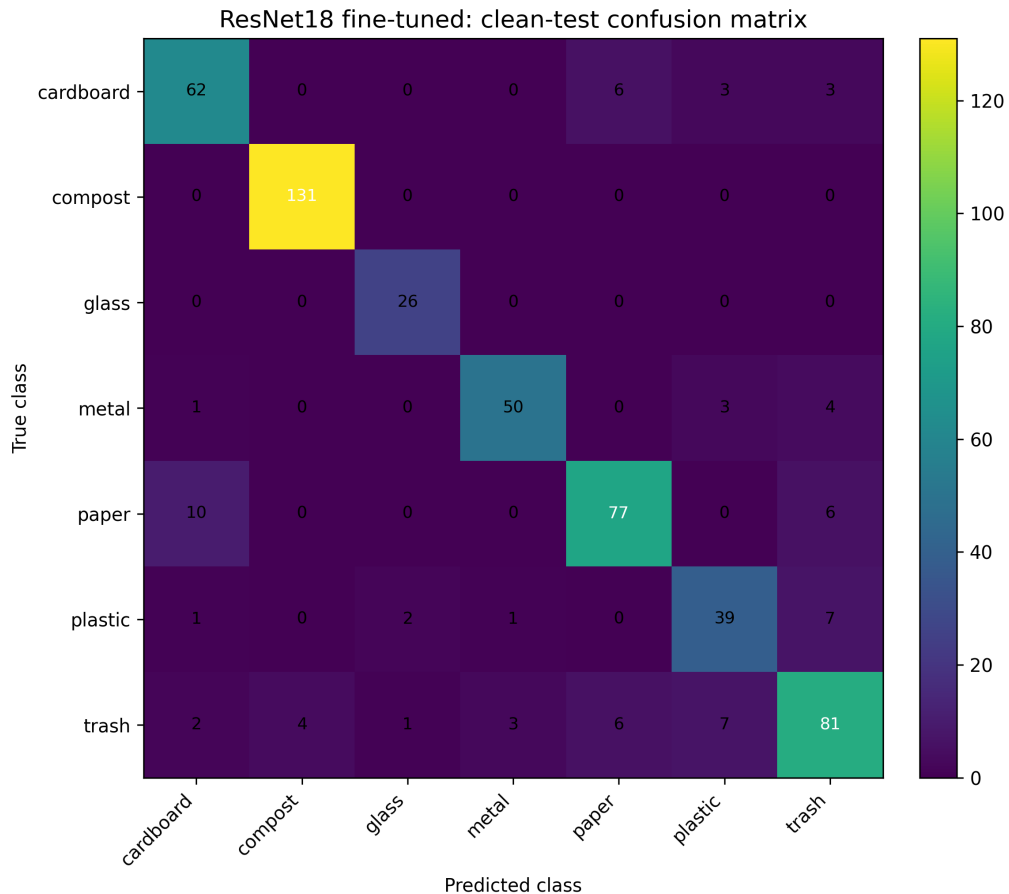


Figure 4. Confusion matrix of ResNet18 fine-tuned on the leakage-free test set.

The diagonal structure of the confusion matrix confirms strong overall discrimination. However, the matrix also makes the remaining errors interpretable. Paper is often confused with cardboard, while plastic and trash are confused in both directions. These patterns are expected because objects in these categories can share texture, shape, material composition, and packaging context.

10. Error analysis

The clean test set contains 70 incorrect predictions out of 536 images. The error-analysis notebook ranks errors by predicted confidence and records the path, true label, predicted label, and confidence for every test instance. This makes it possible to distinguish low-confidence boundary cases from highly confident mistakes.

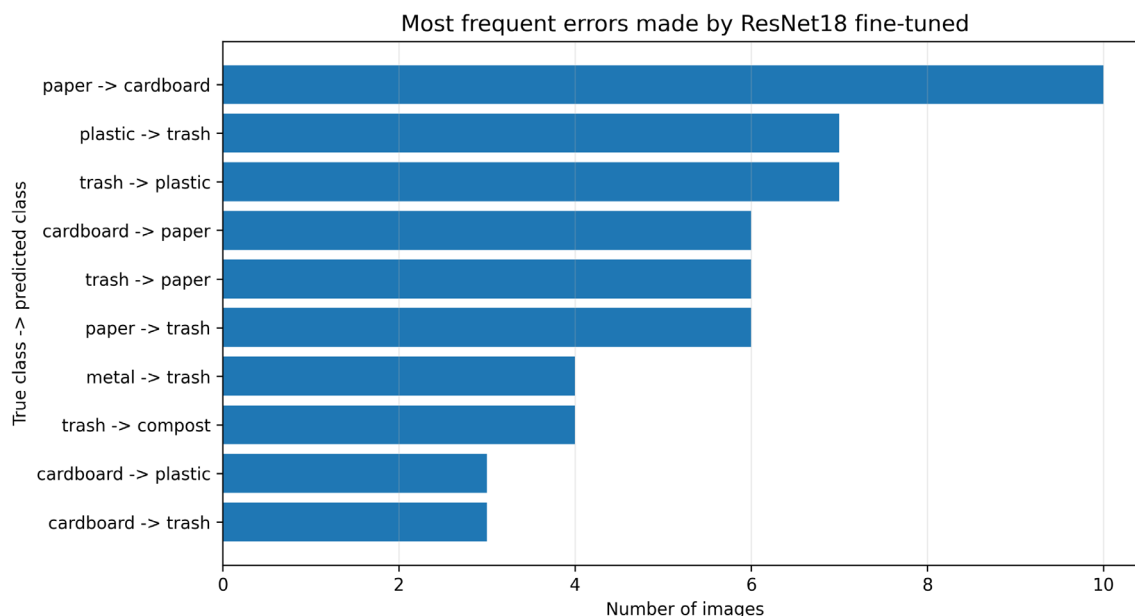


Figure 5. Most frequent true-to-predicted class confusions for ResNet18 fine-tuned.

True class	Predicted class	Count
paper	cardboard	10
plastic	trash	7
trash	plastic	7
cardboard	paper	6
trash	paper	6
paper	trash	6
metal	trash	4
trash	compost	4

The largest error group is paper predicted as cardboard (10 images). This is visually understandable because both classes often contain brown, folded, or sheet-like materials. Plastic predicted as trash and trash predicted as plastic are each observed seven times. Several examples in the qualitative analysis are ambiguous from a recycling perspective: blister packs, cups, composite packages, and product images may combine different materials or depend on local disposal guidance.

Some errors are made with high confidence. This suggests that not all failures are simple uncertainty near a class boundary. They can also reflect label noise, class-definition ambiguity, or image context that differs from the training distribution. This distinction is important in a waste-sorting application because a confident prediction should not automatically be interpreted as an operationally safe recommendation.

11. Discussion

The central finding is that transfer learning is highly effective in this setting. The CNN trained from scratch has limited performance despite augmentation, class weighting, and early stopping. With only 2,088 images available for training after the validation split, learning robust low-level and high-level features from scratch is difficult. The CNN especially struggles with metal, paper, plastic, and glass in the initial evaluation.

By contrast, the frozen ResNet18 obtains strong results while training only a small classification head. This suggests that the pretrained visual filters already capture broadly useful visual information such as edges, textures, shapes, and object configurations. The final fine-tuning step improves both accuracy and macro F1, showing that adapting high-level features to the waste categories is beneficial even when the original pretrained representation is already strong.

The data-integrity audit changes how the results should be interpreted. The initial provided test split gave higher scores, but it contained 108 exact training duplicates. Removing them creates a more credible evaluation protocol. Interestingly, the fine-tuned model still performs strongly after removal, which strengthens confidence that the transfer-learning result reflects genuine generalization rather than only memorization.

The remaining performance gap is concentrated in visually and semantically ambiguous categories. Therefore, future improvements should not focus only on a larger model. Better labels, more realistic images, multi-label annotations, and definitions based on local recycling rules may be as important as architectural changes.

12. Limitations and future work

Limitations

- The dataset is relatively small and class-imbalanced, particularly for glass.
- Some samples appear to be product or stock photos instead of realistic waste scenes, limiting domain realism.
- Waste categories can be ambiguous because objects may be composite, contaminated, or governed by local recycling rules.
- The duplicate audit identifies exact duplicates but not visually near-duplicate images or different crops of the same object.
- The final clean test set is smaller after duplicate removal, so uncertainty around the metrics remains.
- The project evaluates image-level classification only; it does not locate objects, estimate contamination, or support multiple objects in one image.

Future work

- Collect or curate additional real-world waste photographs under varied lighting and backgrounds.
- Use perceptual hashing or embedding-based similarity search to detect near duplicates across all splits.
- Experiment with stronger but controlled augmentation such as color jitter, random erasing, and background variation.
- Use class-specific sampling or focal loss to focus learning on difficult and underrepresented categories.
- Compare ResNet18 with EfficientNet, MobileNet, or vision-transformer backbones under the same clean protocol.
- Move from single-label classification to object detection or multi-label classification for real waste scenes.
- Calibrate probabilities and introduce an abstention option for low-confidence predictions.

13. Conclusion

This project developed a complete and reproducible image-classification workflow for seven waste categories. The workflow included exploratory data analysis, preprocessing, class-imbalance mitigation, a CNN baseline, transfer learning with ResNet18, fine-tuning, quantitative evaluation, and qualitative error analysis.

The main result is clear: the fine-tuned ResNet18 achieves 86.94% accuracy and 86.44% macro F1 on a leakage-free test set of 536 images, compared with 51.68% accuracy and 46.80% macro F1 for the CNN trained from scratch. Pretrained visual representations therefore provide a major advantage for this limited-size dataset. Fine-tuning the final residual stage produces an additional improvement over a frozen backbone.

Equally important, the data-quality audit found 108 exact train/test duplicates in the supplied split. Removing them before final evaluation is a central methodological contribution of this work. It illustrates that high model scores must be interpreted alongside dataset integrity checks. The final project therefore provides not only a strong classifier but also a more trustworthy experimental protocol for evaluating it.

Final takeaway

For small and heterogeneous waste-image datasets, transfer learning with a pre-trained ResNet18 is substantially more effective than training a compact CNN from scratch. However, reliable performance claims require careful dataset auditing, class-aware metrics, and qualitative analysis of errors - not accuracy alone.

References

- [1] rootstrap-org. (2024). Waste Classifier [Dataset]. Hugging Face. <https://huggingface.co/datasets/rootstrap-org/waste-classifier>
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- [5] PyTorch Contributors. (2026). torchvision.models.resnet18 documentation. <https://pytorch.org/vision/stable/models/generated/torchvision.models.resnet18.html>

Submission checklist

- Export this report as PDF and submit it with 15-17 pages excluding optional annexes.
- Keep notebook outputs, figures, and metrics visible.
- Ensure all notebook headings, captions, print statements, and comments are in English.
- Share the repository or provide paths to the four notebooks and the sources used.
- Do not upload the large dataset archive or trained checkpoints to GitHub unless using an appropriate model hub.